

The Interface as a Markov Blanket: Reframing VR Interaction via the Free Energy Principle

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ABSTRACT

This paper argues that VR is a natural “testbed” and “design medium” for the Free Energy Principle and active inference: it can (i) systematically manipulate sensorimotor contingencies, precision weightings, and crossmodal conflicts that shape presence, embodiment, and cybersickness; and (ii) expose interpretable control knobs for interface design in terms of variational and expected free energy. Building on recent VR-FEP perspectives and an active-inference reinterpretation of interaction, we propose a practical mapping from VR components to a Markov-blanket interface, derive design implications for presence and embodied interaction, and outline a measurement-driven agenda for estimating individual sensory characteristics and optimizing VR feedback. The resulting framework reframes usability as the joint reduction of prediction error and complexity and reframes exploration-exploitation in VR as balancing epistemic and pragmatic value.

Index Terms: Virtual reality; active inference; free energy principle; presence; embodiment; sensorimotor contingencies; precision weighting; human-computer interaction.

1 INTRODUCTION

Virtual Reality (VR) has matured from a display technology into a “reality engine” capable of overriding human perception. However, the design of VR interaction remains largely heuristic. We know that increasing field-of-view (FOV) can increase presence, but it can also increase cybersickness, and that visual dominance can override proprioception (as in haptic retargeting). Yet, we lack a unified computational framework to explain why these trade-offs occur and, more importantly, how to optimize them dynamically. Why does the brain accept some virtual hands as “mine” while rejecting others? Why does latency induce sickness in some contexts but not others?

We argue that the *Free Energy Principle (FEP)* and *Active Inference (AIF)* provide this missing link. Beyond its roots in neuroscience, FEP and AIF offer a formal “design language” for HCI. It views the user not merely as an input-processor but as an agent minimizing prediction error through two pathways: changing beliefs (perception) or changing the world (action).

In this view, a VR interface becomes a *Markov Blanket* that mediates the coupling between the user’s internal generative model and the system’s hidden states. This reframing allows us to move beyond notions of “usability” toward a calculable objective: minimizing the user’s Variational Free Energy (F).

This paper proposes that key VR experience and interaction challenges can be productively reframed as problems of *Precision Control*. Specifically, we reinterpret:

- **Presence:** Presence can be interpreted as maintaining low, *resolvable* prediction error under a coherent self-model; many “immersion upgrades” (e.g., wider FOV, lower latency) act as interventions on the precision of sensory evidence rather than ends in themselves.
- **Embodiment:** Embodiment and ownership illusions (e.g., virtual hand ownership) can be viewed as precision-weighted multisensory inference, where visual evidence can dominate proprioception within a tolerance region, enabling techniques such as retargeting and pseudo-haptics.

- **Cybersickness:** Cybersickness can be framed as persistent multisensory prediction errors whose impact is amplified by high precision on conflicting channels; mitigation strategies such as subtle dynamic FOV reduction can be interpreted as context-dependent precision down-regulation [1–3].

We emphasize that our goal is not to replace existing VR theories, but to provide an actionable translation layer. Throughout, each example specifies (i) a user-relevant hidden state x , (ii) the Markov-blanket variables (o, a) exposed by the interface, and (iii) concrete design knobs that modulate precision (signal-to-noise, latency/jitter, crossmodal reliability, and task-driven attention). Here, we use “precision control” as a design-oriented proxy: engineering knobs influence (but do not uniquely determine) the user’s inferred reliability weighting. This framing turns many “rules of thumb” into a shared control problem. Furthermore, FEP provides a single modeling vocabulary for relating “motion sickness”, “embodiment”, and “operability” in terms of (approximate) model evidence and its bounds, without implying a single engineered controller that optimizes all objectives.

This work translates the abstract mathematics of FEP into actionable design knobs for the next generation of adaptive VR systems.

2 OVERVIEW OF FREE ENERGY PRINCIPLE

2.1 Prediction, inference, and action

In predictive coding accounts, the brain is modeled as a hierarchical generative model that continuously generates top-down predictions of sensory input and updates beliefs when prediction errors arise [4]. A core idea is that perception is not passive “signal processing,” but approximate Bayesian inference: latent causes (hidden states) are adjusted so that predicted sensations match incoming observations. In many formulations, only *prediction errors* are propagated bottom-up, while predictions flow top-down.

A key ingredient is *precision* (informally, the estimated reliability of a signal; often treated as an inverse variance). Precision controls the gain of prediction errors and therefore determines which modalities or features dominate belief updating. This precision-weighting is frequently interpreted as an attentional or context-setting mechanism: the same sensory discrepancy can be ignored as noise or

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treated as decisive evidence depending on its inferred precision.

The FEP provides a broader rationale for why prediction-error minimization matters. In brief, systems that persist must avoid repeatedly visiting highly surprising sensory states; however, surprise $-\ln p(o)$ is generally intractable to evaluate directly. FEP therefore introduces *variational free energy* as a computable quantity whose minimization indirectly bounds (and thus reduces) surprise [5]. Predictive coding can be understood as one class of message-passing/gradient-descent schemes that minimize this variational objective under simplifying assumptions.

Active inference extends predictive coding by emphasizing that agents can reduce prediction error not only by updating beliefs (perceptual inference) but also by acting to sample observations that fulfill predictions (active inference) [5, 6]. Intuitively, when the world violates expectations, an agent can either change its model of the world or change the world (via action) so that sensations conform to its model. This coupling between inference and action makes the framework especially natural for VR, where systems can explicitly manipulate sensorimotor contingencies and the apparent reliability (precision) of feedback channels.

2.2 Variational free energy as an accuracy–complexity trade-off

Let o denote observations and x hidden states. Variational free energy F can be defined as

$$F(o, q) = \mathbb{E}_{q(x)}[\ln q(x) - \ln p(o, x)],$$

and, assuming $p(o, x) = p(o | x)p(x)$, it can be written as the familiar accuracy–complexity decomposition:

$$F(o, q) = \underbrace{D_{\text{KL}}[q(x) \| p(x)]}_{\text{complexity}} - \underbrace{\mathbb{E}_{q(x)}[\ln p(o | x)]}_{\text{accuracy}}. \quad (1)$$

The first term is a complexity cost (how far posterior beliefs move from priors); the second is an accuracy term (how well beliefs explain current observations).

Although complexity is formally the KL divergence between posterior beliefs and priors, it provides a useful proxy for the cost of revising internal models. In VR, hidden mode switches, inconsistent mappings, or abrupt remappings can increase this cost even when sensory fidelity is high; conversely, clear feedback and stable mappings improve accuracy without forcing costly belief updates. This is the sense in which usability improvements can be described as shaping the user’s accuracy–complexity trade-off.

2.3 Expected free energy and exploration–exploitation

Future-oriented action selection is often described by minimizing expected free energy $G(\pi)$ under a policy π [6]. Operationally, $G(\pi)$ evaluates the consequences of a candidate policy in terms of both information-seeking and goal-directed value. A useful decomposition highlights two values:

$$G(\pi) \approx \underbrace{-\mathbb{E}[\text{information gain}]}_{\text{epistemic value}} - \underbrace{\mathbb{E}[\ln p(o | C)]}_{\text{pragmatic value}}. \quad (2)$$

Equation 2 is schematic; we use it as an intuitive decomposition rather than a full generative-model specification. Epistemic value favors actions that reduce uncertainty (e.g., probing to learn mappings); pragmatic value favors outcomes aligned with preferences C (e.g., completing a task with minimal cost). In VR, this suggests a unified account of “learning the world” (exploration) and “getting things done” (exploitation), and it motivates adaptive interfaces that manage precision and feedback to support both phases without destabilizing user priors.

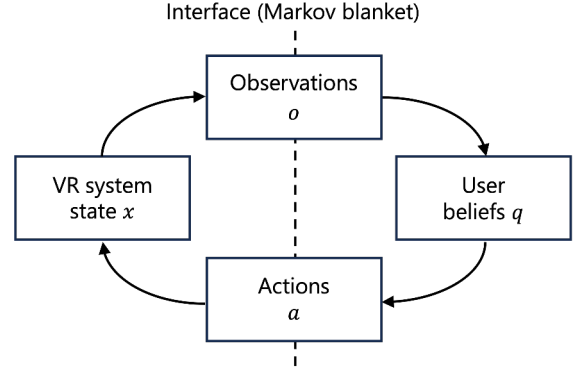


Figure 1: A Markov-blanket view of VR: the interface (actions and observations) mediates coupling between user beliefs and hidden system state.

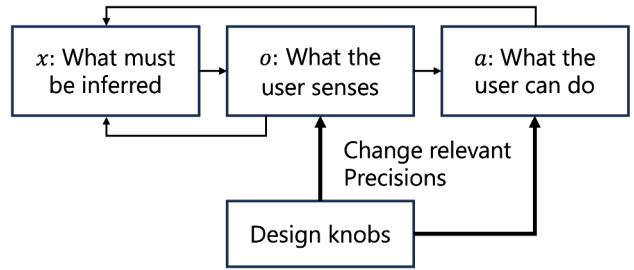


Figure 2: A worked example template for mapping a VR scenario to hidden states x , observations o , and actions a (Markov-blanket variables), and to corresponding precision-control levers (instantiated in Sec. 4.1–4.3).

3 REFRAMING VR INTERACTION AS FREE-ENERGY MINIMIZATION

3.1 The interface as a Markov blanket

A Markov blanket separates an agent’s internal states from external states via sensory and active states. We use *Markov blanket* as a modeling analogy: actions and observations are mediating variables that screen user beliefs from the system’s hidden state. Figure 1 visualizes this Markov-blanket interface view. For VR/HCI, we can view the user as an inference agent whose internal states (beliefs, intentions) are coupled to the system’s hidden states (simulation variables, UI state) only through: (i) *sensory states*—rendered pixels, audio, haptics; and (ii) *active states*—controller motion, gaze, voice, body tracking. This abstraction emphasizes that “looking at a display” is better described as inferring hidden system states from observations and then acting to request new observations.

Crucially, VR systems can shape both sides of this coupling. They can modify the observation model (what evidence is presented and with what temporal/spatial noise) and the action model (how movements map to consequences via gains, constraints, and prediction). This makes VR a particularly direct medium for implementing and testing precision-control interventions that would be difficult to realize in the physical world.

3.2 Usability and UX as objective shaping

Within Equation 1, usability improvements can be expressed as (a) increasing accuracy by making system state easier to infer (clear mappings, informative feedback), and/or (b) reducing complexity by keeping priors stable and minimizing unnecessary model revisions. This reframes the “cost of learning” and “surprise” as quantities that design can control.

In many VR/HCI experiences, users alternate between learning how interaction works (high epistemic value) and executing well-learned actions (high pragmatic value). Design elements such as tooltips, previews, tutorials, or gradual disclosure can be interpreted as providing epistemic resources that accelerate belief formation and reduce long-run free energy.

4 IMPLICATIONS FOR PRESENCE, EMBODIMENT, AND CYBERSICKNESS

Figure 2 is a reusable template for translating a VR phenomenon into active-inference terms. In each subsection below, we instantiate the same Markov-blanket variables—hidden states x , observations o , and actions a —and then name the concrete design knobs that primarily modulate (i) likelihood/temporal precision and (ii) policy/actionability. Sections 4.1–4.3 apply this template to presence (resolvable prediction error), embodiment (precision-weighted visuo-proprioceptive fusion), and cybersickness (persistent conflict requiring policy change or precision down-regulation), respectively.

4.1 Presence as controlled prediction error

Mapping x : scene and self-state (“what is happening now, and what am I doing?”); **o :** visual/auditory feedback plus interoceptive comfort cues; **a :** head/hand/body movements (incl. pause/recenter); Design knobs: temporal precision (latency/jitter), cue reliability (likelihood precision), and task-driven attention.

Presence has both subjective (phenomenal) and functional interpretations, and has been discussed as “being here now” and as action-capability in the virtual world [7, 8]. From an active-inference lens, presence corresponds to maintaining low prediction error across exteroceptive and interoceptive channels under a coherent self-model. VR can deliberately tune this by aligning sensorimotor contingencies and by managing precision weights. This is compatible with broader accounts that connect interoceptive inference to sensorimotor contingencies in the “cybernetic Bayesian brain” [9].

Mini-case (latency vs. presence). Consider a reach-to-grasp interaction in which the user infers whether their action has caused a virtual object to move. Here, increasing immersion (lower end-to-end latency, higher visual fidelity) effectively increases visual precision ω_{vision} , improving accuracy and strengthening agency. However, higher precision also increases the salience of temporal mismatch (e.g., delayed haptics or late visual updates), which can push errors from “resolvable” to “irreducible.” A practical implication is to either reduce mismatch (engineering) or down-regulate the unreliable channel’s precision (e.g., soften haptic onset, avoid ultra-crisp contact cues) during periods when the system cannot guarantee tight sensorimotor consistency.

4.2 Embodiment as precision control over proprioception

Mapping x : body state and avatar correspondence (pose/contact; ownership/agency hypotheses); **o :** vision, proprioception, touch; **a :** voluntary movement and exploratory “probing”; Design knob: visuo-motor alignment/delay, haptic reliability/strength, retargeting magnitude, and instructions (precision re-weighting via attention).

The sense of embodiment in VR commonly includes agency, body ownership, and self-location [10]. Active inference models describe bodily selfhood as a hierarchical generative model that predicts the sensory consequences of action [11, 12]. A key mechanism is precision control: attenuating proprioceptive prediction errors can facilitate adapting body models to novel visual feedback [13]. This perspective naturally links VR illusions (e.g., virtual hand ownership) to top-down modulation of sensory precision.

4.2.1 Crossmodal interfaces and retargeting as engineered priors

Crossmodal techniques such as pseudo-haptics [14] and haptic retargeting [15] can be interpreted as interventions on the evidence model: they reshape the mapping from hidden states to observations to “explain away” missing hardware channels. In active-inference terms, they work when induced prediction errors fall within a tolerance region where the agent can resolve conflict via belief updating rather than rejecting the experience.

Mini-case (retargeting vs. haptic realism). In passive-haptics setups, designers often add stronger tactile cues to increase realism. The precision-control view predicts an interaction: increasing haptic/proprioceptive reliability raises ω_{proprio} , shrinking the “tolerance region” in which visual capture can explain away geometric discrepancies. Practically, this suggests co-design: either (i) reduce retargeting error when adding strong haptics, or (ii) provide epistemic scaffolding (calibration and stable mappings) so users can update priors without accumulating surprise.

4.3 Cybersickness as unresolved conflict under high precision

Mapping x : self-motion and expected sensorimotor contingencies; **o :** optic flow (especially peripheral), vestibular cues, proprioception; **a :** locomotion control and corrective actions (stop/recenter/teleport); Design knob: temporal mismatch (latency/jitter), gain/acceleration, locomotion policy (smooth vs. teleport), and visual precision shaping (vignette/blur).

Cybersickness arises when visual, vestibular, and proprioceptive cues conflict and is commonly quantified with standardized instruments such as the Simulator Sickness Questionnaire (SSQ) [1]. Rather than replacing established VR accounts (e.g., sensory-conflict and postural-instability perspectives) [3, 16, 17], we propose a computational translation: cybersickness corresponds to persistent prediction errors that cannot be reduced by action or belief updating under current sensorimotor constraints, with their impact amplified by high precision on conflicting channels. This framing motivates concrete levers—reduce delay and jitter, avoid inconsistent acceleration, and apply context-dependent precision shaping (e.g., smoothing or vignetting) when users cannot act to resolve the conflict [2].

Mini-case (smooth locomotion and precision gating). Consider joystick-based smooth locomotion while the user remains physically stationary. High-fidelity optic flow provides strong evidence for self-motion, yet vestibular and postural cues predict stasis, yielding a conflict that cannot be fully resolved by action. Two complementary responses follow: (i) increase *actionability* by offering immediate corrective options (stop/recenter/teleport) and by limiting unexpected acceleration, and (ii) when actionability is low (e.g., on-rails motion), temporarily down-regulate the precision of peripheral optic flow via subtle vignetting during high-speed/high-acceleration intervals [2]. This framing predicts that always-on vignetting is sub-optimal: it should be gated by conflict proxies (speed/acceleration, head-motion stability) and disengaged when users switch to a low-conflict policy such as teleportation.

5 VR AS A TESTBED FOR ACTIVE INFERENCE

VR is not only a target application; it can also serve as an experimental platform for testing active-inference hypotheses. Compared to real-world experiments, VR offers precise control over observation models (rendering, audio, haptics) and action mappings (tracking, gains, delays). VR enables a testbed with the following FEP-inspired manipulations.

Table 1: Reinterpreting VR Design Knobs through Active Inference: From Heuristics to Precision Control (here, ω denotes inferred precision, approximately inverse variance, that weights prediction errors).

VR Design Technique	Standard UX Heuristic	Active Inference Interpretation (Optimization Logic)
Dynamic FOV Reduction (e.g., vignetting during locomotion) [2]	Reduces optical flow to prevent motion sickness (often at the cost of peripheral detail and presence).	Precision Down-regulation ($\downarrow \omega_{vision}$): Artificially reduces the precision of visual evidence during high-conflict states (sensory mismatch), preventing the user’s model from accumulating irreducible prediction errors that trigger sickness.
Haptic Retargeting (Hand redirection) [15]	Exploits “Visual Capture” to override proprioception. This can fail when discrepancies exceed detection thresholds or task demands increase.	Precision Weighted Fusion: Exploits the generative model’s property where posterior beliefs follow the highest-precision channel. The system relies on $\omega_{vision} \gg \omega_{proprio}$ to update the belief of hand location without updating the belief of body structure (complexity).
Adaptive UI / Guidance	Helps the user but can be annoying if intrusive.	Minimizing Expected Free Energy (G): The system selects interventions that maximize Pragmatic Value (goal achievement) while strictly bounding the Complexity cost (divergence from the user’s learned policy/priors).

5.1 Precision manipulation via attention and task context

Precision weighting—how strongly prediction errors update beliefs—is often linked to attention. VR enables controlled shifts in attentional set by changing task demands, modality visibility, or modality reliability. For example, visuo-proprioceptive conflicts can be introduced by delaying a virtual hand relative to the real hand and then instructing participants to match either the virtual or real movement [18]. Such paradigms provide a direct route to estimate modality-specific precisions and to test whether attention reallocates precision between vision and proprioception.

5.2 Measurement-driven personalization

A complementary direction is to treat personalization as an inference problem: estimate user-specific precision weightings and use them to set rendering/haptics parameters. Sensory characteristics vary widely across users, and clinical work operationalizes “sensory profiles” via thresholds and behavioral responses [19]. A lightweight VR calibration could apply controlled perturbations—(i) visuomotor delay, (ii) visual gain/offset, (iii) vibrotactile reliability, and (iv) FOV reduction—while measuring adaptation curves, exploratory “probing” behavior, task performance, and SSQ. A simple active-inference model can then treat these precisions as latent variables; identifiability can be improved via blocked/factorial manipulations and hierarchical priors across tasks. The resulting static “sensory profile” can parameterize runtime settings without requiring heavy real-time inference, aiming to reduce chronic conflict while preserving epistemic value early in learning.

5.3 Two-Sided Active Inference for Adaptive VR

The most consequential implication for VR systems is that the *system* can also be modeled as an inference agent. An adaptive VR interface can infer latent user intent and uncertainty from interaction traces, and then choose interventions (hints, haptics, guidance) that reduce expected free energy.

Let u be a latent user goal and y observed interaction features (gaze, motion, timing). The system maintains $q(u | y)$ and selects an assistance action α by minimizing a surrogate of $G(\pi)$ in Equation 2

that trades off: (i) pragmatic value (task progress), (ii) epistemic value (disambiguating intent), and (iii) user complexity cost (surprise induced by intervention). This suggests that even simple adaptive behaviors—e.g., only showing a clarification prompt when uncertainty is high, or only snapping when confidence is high—can be justified in a unified objective.

6 DISCUSSION AND LIMITATIONS

Two caveats matter. First, “free energy” is a theoretical construct, not a directly measurable signal. Practical work must use proxies (e.g., adaptation dynamics, gaze, pupil, EEG markers, self-report) and validate their mapping to model variables. Second, system-side adaptation can backfire: overly dynamic interfaces can violate user priors, increasing complexity and harming usability. Active inference suggests an explicit control problem: intervene only when the system’s posterior confidence about the user’s intent (intent precision, not sensory precision) is high.

7 CONCLUSION

We presented a concise bridge between active inference and VR. By viewing the interface as a Markov blanket, interaction becomes a coupled process of belief updating and information-seeking action. This view yields practical design questions—how to shape accuracy vs. complexity costs, and how to balance epistemic vs. pragmatic value—and it suggests that core VR phenomena (presence, embodiment, and cybersickness) can be approached as problems of precision control. We also argued that VR is a uniquely controllable platform for testing active-inference hypotheses and for estimating individual sensory characteristics to personalize feedback. Future work should connect computational variables to robust behavioral and physiological proxies, and explore “two-sided” active inference where human and system mutually adapt without destabilizing user priors.

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